

September 29 Wing Lecture Guide

Instructor run of show | MIE 446 | 11:30 a.m. to 12:45 p.m.

Finish the wing portion of Lecture 01, then move each assigned team from structural vocabulary to a baseline layout and an R01 coupon file. Do not spend this meeting deriving every wing-load equation.

11:30-11:36	Lecture 01 section 7 Say and show: "Last class we identified tail surfaces. Today, where does a load on a wing go?" Draw a half-wing next to the tail sketch. Students: Mark root, tip, leading edge, and trailing edge.
11:36-11:50	Sections 7A and 7A-1 Say and show: Project the FAA wing cutaway and the wing-structure figure. Point to skin, rib, spar, stringer, and wing box; connect each name to a job. Students: Label shell, ribs, two rod paths, and root on view A.
11:50-12:00	Sections 7A-2 to 7A-4 Say and show: Show flap, aileron, spoiler, slat, and the spoileron versus wing-tip-rudder comparison. A control role does not identify identical hardware. Students: Identify the real-wing devices; exclude them from the printed wing.
12:00-12:10	Sections 7A-5, 7B and 7C Say and show: Trace surface load to root. Cut the outboard wing and introduce V_z , M_x , and possible T_y by meaning only; full calculation follows later. Students: Draw one load arrow, a path to root, and a cut with V_z , M_x , T_y .
12:10-12:30	Project layout studio Say and show: State the fixed 450/160/100 mm baseline, 3 modules, two 4 mm rods. A student wing is a non-flying demonstration model. Students: Complete both sketches, dimension checks, and a prediction.
12:30-12:43	Wing Designer steps 1-4 Say and show: Open a prepared Colab copy. Show Coupon Only and revision R01. Never enter a coupon result before a physical fit test. Students: Save a team copy; run R01 and note the ZIP or a blocker.
12:43-12:45	Exit check Say and show: Ask: "What do we have, and what do we still not have?" Students: Answer: layout and R01 file; physical fit, R02 CAD, and print authorization are still pending.

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Instructor checks and preparation | Keep this page off the student handout

Baseline inputs to project on screen

Workflow and revision	Coupon Only; R01; real team ID and names
Airfoil and geometry	NACA 2412 unless another section is approved; $s = 450$ mm; root/tip chords = 160/100 mm
Construction	Skin = 1.2 mm; rib = 1.6 mm; 3 modules; two fixed 4 mm rod paths at $x/c = 0.30$ and 0.60
Comparison only	Set comparison tip chord to 90 mm; predict area down, aspect ratio up; production tip chord stays 100 mm
Coupon state	COUPON_CONFIRMED = False; fit result = Not tested yet

Numbers for a quick desk check

Module seams	$y = 150$ and 300 mm; nominal segment length = 150 mm
Chord at seams	$c(150) = 140$ mm; $c(300) = 120$ mm
Root rod centers	$x = 48$ and 96 mm aft of the local leading edge
Tip rod centers	$x = 30$ and 60 mm aft of the local leading edge
October 1 geometry	Half area = 58,500 mm ² ; full area = 0.1170 m ² ; taper = 0.625; equivalent full-wing AR = 6.923

Before students arrive

Open Lecture 01 at section 7A and pre-run its setup and figure cell. Open one tested copy of Wing Designer. Confirm students have the worksheet link, Colab access, assigned team IDs, a course rod, and staff-supervised coupon appointments. The class cannot physically print 21 coupons in a single thirteen-minute notebook block.

After-class handoff

Collect the R01 sheet and coupon ZIP or a specific blocker. Move coupon printing and rod-fit tests into authorized appointments before the October 6 Final Wing run. A missing physical fit record must hold the team at Coupon Only.